Prototype bring-up

For: the person at the bench with the assembled prototype board. **You need:** - a bench supply with an adjustable current limit, 0–20 V - a multimeter - a 100 MHz oscilloscope with a probe rated for 100 V - an ST-LINK V3 or J-Link with the 10-pin 1.27 mm Cortex cable - dummy loads for steps 4 and 5

Power the board up **one supply at a time**, in the order below. Each supply has its own terminal: pin 21 is the logic feed, pin 04 the injector feed, pin 06 the EGR feed. So each block comes up with everything after it simply unpowered. Don't skip a step. If a reading is outside its range, **stop** and find out why before going on.

Expected values come from the production simulation blocks (`sim/blocks/`) and the sheet notes; each is named.

Where things are

Terminal	Edge	Position: signal (production pin)
J1001	left, top	1: VBAT_IN (21) · 2: V_BAT_1R (04, injector) · 3: V_BAT_2R (06, EGR) · 4: GND
J1002	left	1, 2: GND, the power return
J1003	left	injectors: INJ_A_03, INJ_B_05, INJ_LS1_73, INJ_LS2_29, INJ_LS3_07
J1004	top	MU_RETURN (88), EGR_HIGH_59, EGR_LOW_81, RLY_MAIN_OUT (50), RLY_BUZZER_OUT (69)
J1005	right	sensors: 5 excitations, 6 sensor inputs, SGND 34/36, GND (rail sensor return, 08)
J1006	right	speed: CRANK_HI_52, CRANK_LO_74, CAM_FREQ_46, CAM_GND_44, SHIELD_30, GND
J1007	bottom	switches 43/23/20/24/71, TRIP_LOOP
J1008	bottom	CAN1_H, CAN1_L, GND, CAN2_H, CAN2_L, GND
J1009	inside, by J1005	2 × 10 bridge header: crank pair, cam, 6 sensor inputs, ignition on the odd pins; GND on the even pins
J301	by the MCU	SWD (Cortex 10-pin). SWO is wired to pin 6
J302	by the MCU	USB-C, **data only**: it does not power the board

Test points (TP1–TP24):

TP	Net	TP	Net
TP1	VBAT_PROT	TP13, TP14	GND (injector)
TP2	VBAT_INJ	TP15, TP16	ISNS_INJ_A, ISNS_INJ_B
TP3	VBAT_EGR	TP17, TP18	ISNS_MU, ISNS_EGR
TP4	12V_GATE	TP19, TP20	CRANK_EDGE, CAM_EDGE
TP5	5V_MAIN	TP21, TP22	CAN1_H, CAN1_L

TP	Net	TP	Net
TP6, TP11	GND	TP23, TP24	CAN2_H, CAN2_L
TP7	3V3_MCU	**TP12**	**BOOST_100V: up to 116 V. HV.**
TP8–TP10	5V_SENSOR_A/B/C		

1. Before any power

1. **Look** at every IC for orientation and solder bridges, under a loupe. Check the LQFP-100 (U301) and the HVSSOP boost controller (U401) closely.
2. **Resistance to GND** with the meter. Nothing on this list should read below 100 Ω. A short here would destroy parts at power-up.

Probe	Expect
TP1, TP2, TP3 (the feeds)	> 1 kΩ
TP5 (5V_MAIN), TP7 (3V3_MCU)	> 100 Ω (charging capacitors: the reading climbs)
TP12 (BOOST_100V)	> 10 kΩ, climbing (two 82 µF capacitors charging)
TP21–TP22, TP23–TP24 (H to L of each bus)	**120 Ω**, the split termination 60R0 + 60R0 (`can` sheet)

2. Logic only: pin 21, limited to 100 mA

Supply **13.5 V on J1001-1**, with GND on J1002. Leave pins 04 and 06 open.

Check	Expect	Source
Supply current	under 100 mA. **If it sits at the limit, switch off**	—
TP1 VBAT_PROT	about 13.5 V, less one FET drop	`reverse_battery.cir`
TP5 5V_MAIN	5.0 V, ripple under 50 mV p-p	`buck_preregulator.cir`
TP7 3V3_MCU	3.30 V. **It must sit inside 3.143–3.459 V**, or the supervisor resets the board. The TLV767 holds 1 %	`rails` sheet; `supervisor.cir`
TP8–TP10 5V_SENSOR_A/B/C	about 5.0 V, following 5V_MAIN	`sensor_rail.cir`
D302 (3V3 LED)	on	—
TP4 12V_GATE	**about 0 V**. It comes from the boost rail, which is unpowered	`injector` sheet

Then connect the debugger to J301. **SWD must connect,** and the debugger should read the device ID (STM32G474, `0x469`). Flash a test build that toggles `LED_STATUS` (PB7): D303 should blink.

The supervisor will reset the MCU every few tens of milliseconds unless the firmware kicks `WDT_KICK` (PC11) at 5–20 ms intervals (decisions log 11). With no firmware, a reset pulse train on NRST is correct behaviour, not a fault.

3. CAN and the sensor inputs

1. **CAN:** connect a USB-CAN adapter to J1008 at 250 kbit/s. Each bus is split-terminated on the board; terminate the adapter end too. Send and receive a frame on each bus.
2. **Sensors:** put a 10 kΩ potentiometer between each excitation and its SGND on J1005, with the wiper on the matching input. Sweep each one and read the ADC. - **The divided channels read about 0–3.07 V for 0–5 V in.** `sensor_ratiometric.cir` checks 5.0 V at the pad giving 3.072 V.
3. **Battery monitor:** `VBAT_MON` (PB12) should read **1.04 V at 13.5 V** (120 k / 10 k).

4. The feeds: pin 06 (EGR), then pin 04 (injector), limited to 1 A

Pin 06 first: connect 13.5 V to J1001-3 at a 1 A limit. - TP3 VBAT_EGR should read about 13.5 V. - Load the EGR outputs (J1004-2/-3) with a 10 Ω, 25 W resistor. - The EGR bridge stays coasting until firmware raises `EGR_EN`.

Then pin 04: connect 13.5 V to J1001-2 at a 1 A limit. Put a dummy load on each injector output in J1003: 0.2 Ω + 200 µH, or a real injector on a bench.

Check	Expect	Source
TP2 VBAT_INJ	about 13.5 V	`power_path.cir`
TP12 BOOST_100V (HV probe)	**99.7 V nominal**, never above about 102 V	`boost_converter.cir` ; `injector` sheet
`BOOST_MON` (PB11)	**2.43 V at 100 V** (24.9 k / (2 × 499 k + 24.9 k) = 0.0243)	`prototype/additions.py`
TP4 12V_GATE	about 12.3 V (a 13 V zener-referenced follower from the boost rail). The drivers' UVLO is 9.8 V	`injector` sheet

With firmware firing one injection at 1 ms: - **TP15/TP16** (ISNS_INJ) should show the current rising to the 18 A peak in about **38 µs**, from the boost rail (`injector_boost.cir`). - After turn-off, the current should fall in about **36 µs** (`injector_turnoff.cir`). - **BOOST_100V** should be back at its setpoint before the next cylinder, 26.67 ms later.

5. The watchdog-kill test

1. With pins 21, 04 and 06 powered and the firmware driving the outputs, **stop kicking** `WDT_KICK`.
2. Within the watchdog window, the TPS3850 asserts RESET. Q301 turns off, and `GATE_KILL` rises to about **12 V** (zener-clamped, 12.7 V maximum). Probe it at R305/Q301's drain.
3. **Every gate must drop:** - injector high and low sides - metering FET - EGR bridge - relays

Check the current on each dummy load, and the gate of Q412 (low side 1) on the scope. 4. `WDT_FAULT` (PC12) reads low after the reset. This is how firmware tells a watchdog trip from a rail trip. **Since 15 Oct 2026 this is a separate net from RESET:** they were shorted together on the production schematic until this prototype found it.

Pass: all five steps within their ranges. Record the readings in `docs/status.md` with the date and board serial.